

Tutorial Sheet 2

Solutions

Question (1)

(a) $x \in \left[1 - \sqrt{2}, \frac{1}{2}\right) \cup [1 + \sqrt{2}, \infty)$

(b*) $x \in \left[-3, \frac{1}{2}(-3 + \sqrt{5})\right)$

(c) $x \in (-1, 1)$

Note: To correctly solve case (b), consider the following:

- (i) The RHS $\sqrt{x+3}$ is defined only for $x \geq -3$. Solutions of the given inequality must thus always be larger than or equal to -3 .
- (ii) Given $x \geq -3$, if $x + 2 < 0$ then the inequality is automatically verified, since the square root on the RHS is always positive or zero. So $-3 \leq x < -2$ is always a solution.

After these two conditions are considered, squaring both terms provides us with the final result.

Question (2)

(a) Let us verify if the property $\mathcal{P}(n) : (n+1)^2 \geq n^n$ is true for $n = 1, \dots, 4$:

$\mathcal{P}(1)$	:	$(1+1)^2 = 4, 1^1 = 1$	$\Rightarrow$	$4 \geq 1$	✓
$\mathcal{P}(2)$	:	$(1+2)^2 = 9, 2^2 = 4$	$\Rightarrow$	$9 \geq 4$	✓
$\mathcal{P}(3)$	:	$(1+3)^2 = 16, 3^3 = 27$	$\Rightarrow$	$16 \not\geq 27$	✗
$\mathcal{P}(4)$	:	$(1+4)^2 = 25, 4^4 = 256$	$\Rightarrow$	$25 \not\geq 256$	✗

The property is not true for any integer larger than 2. In fact, for any $n \geq 3$, $n^n > (n+1)^2$. As we will see, n^n grows very fast as n increases.

(b) Similarly, for $\mathcal{P}(n) : n! \geq n^n$, we have

$$\begin{array}{llll}
 \mathcal{P}(1) & : & 1! = 1, 1^1 = 1 & \Rightarrow 1 \geq 1 \quad \checkmark \\
 \mathcal{P}(2) & : & 2! = 1 \cdot 2 = 2, 2^2 = 4 & \Rightarrow 2 \not\geq 4 \quad \times \\
 \mathcal{P}(3) & : & 3! = 1 \cdot 2 \cdot 3 = 6, 3^3 = 27 & \Rightarrow 6 \not\geq 27 \quad \times \\
 \mathcal{P}(4) & : & 4! = 24, 4^4 = 256 & \Rightarrow 24 \not\geq 256 \quad \times
 \end{array}$$

The property is not true for any integer larger than 1.

(c) Finally, for $\mathcal{P}(n) : \ln^2(1+n) \geq (n-2)$, we will eventually need to use the calculator to compute values, however we can address the first three without a calculator:

$$\begin{array}{llll}
 \mathcal{P}(1) & : & (\ln 2)^2 > 0, (1-2) = -1 & \Rightarrow LHS \geq RHS \quad \checkmark \\
 \mathcal{P}(2) & : & (\ln 3)^2 > 0, (2-2) = 0 & \Rightarrow LHS \geq RHS \quad \checkmark \\
 \mathcal{P}(3) & : & (\ln 4)^2 > 1 \text{ since } e < 4, (3-2) = 1 & \Rightarrow LHS \geq RHS \quad \checkmark \\
 \mathcal{P}(4) & : & (\ln 5)^2 \approx 2.59, (4-2) = 2 & \Rightarrow LHS \geq RHS \quad \checkmark
 \end{array}$$

The property is true for $n = 1, \dots, 4$. The property is, however, false for $n \geq 6$, since $(\ln 6)^2 \approx 3.78 < (6-2) = 4$.

Question (3)

Part (a) is very similar to question (2a) in Section 1.2 “The induction principle” of the questions booklet for the course. We refer the student to that proof.

Part (b) is identical to question (1b) in Section 1.2 of the question booklet. To prove it, we show that

$$\mathcal{P}(n) : 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

is valid for $n = 1$. We have:

$$LHS = \mathcal{P}(1) = 1^2 = 1, \quad RHS = \frac{1 \cdot (1+1) \cdot (2 \cdot 1 + 1)}{6} = \frac{1 \cdot 2 \cdot 3}{6} = 1 = LHS.$$

Then, we state $\mathcal{P}(n+1)$, using the expression for $\mathcal{P}(n)$. On the LHS we add the extra term coming from the sum, i.e. $(n+1)^2$. On the RHS we replace n with $(n+1)$ everywhere. We have:

$$\begin{aligned}
 \mathcal{P}(n+1) & : \quad \overbrace{1^2 + 2^2 + 3^2 + \dots + n^2}^{\text{LHS of } \mathcal{P}(n)} + (n+1)^2, \quad \overbrace{\frac{(n+1)((n+1)+1)(2(n+1)+1)}{6}}^{\text{RHS of } \mathcal{P}(n): n \rightarrow (n+1)} \\
 & = \frac{(n+1)(n+2)(2n+3)}{6}
 \end{aligned}$$

We will need to prove that these are identical.

Now we use the assumption that $\mathcal{P}(n)$ is true. This implies that the LHS of $\mathcal{P}(n+1)$ can be rewritten using $\mathcal{P}(n)$, because it contains the sum of the squares of the first n numbers:

$$\overbrace{1^2 + 2^2 + 3^2 + \dots + n^2}^{\text{RHS of } \mathcal{P}(n): \frac{n(n+1)(2n+1)}{6}} + (n+1)^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2.$$

Expanding the RHS explicitly, we find:

$$\frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{(n^2+n)(2n+1) + 6(n+1)^2}{6} = \frac{2n^3 + 9n^2 + 13n + 6}{6}$$

This can now be proven to be equal to the RHS of $\mathcal{P}(n+1)$. In fact:

$$\frac{(n+1)(n+2)(2n+3)}{6} = \frac{(n^2+3n+2)(2n+3)}{6} = \frac{2n^3 + 9n^2 + 13n + 6}{6}$$

In conclusion, we have proven that the LHS of $\mathcal{P}(n+1)$ is equal to the RHS of $\mathcal{P}(n+1)$, i.e. that $\mathcal{P}(n+1)$ is true if $\mathcal{P}(n)$ is true. The induction theorem allows us to state that the property is thus valid for all numbers $n \in \mathbb{N}$.

Question (4)

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|---|---|
| (a) $D = \left[-\frac{1}{2}, \sqrt{3}\right) \cup (\sqrt{3}, \infty)$ | (b) $D = (-\infty, 1] \cup [4, \infty)$ |
| (c) $D = \mathbb{R}$ | (d) $D = \mathbb{R}$ |
| (e) $D = \mathbb{R} \setminus \{1\}$ | (f) $D = \mathbb{R}$ |

Question (5)

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|---|--|
| (a1) $f \circ g = f(g(x)) = \frac{1}{\sqrt{\ln(x^2+3)}+1}$ | $D_{f \circ g} = \mathbb{R}$ |
| (a2) $g \circ f = g(f(x)) = \ln \left(\left(\frac{1}{\sqrt{x}+1} \right)^2 + 3 \right)$ | $D_{g \circ f} = [0, \infty)$ |
| (b1) $f \circ g = f(g(x)) = \exp \left(\frac{x^2+1}{x^2-1} \right)$ | $D_{f \circ g} = \mathbb{R} \setminus \{-1, 1\}$ |
| (b2) $g \circ f = g(f(x)) = \frac{e^{2x}+1}{e^{-2x}-1}$ | $D_{g \circ f} = \mathbb{R} \setminus \{0\}$ |

Question (6)

We indicate the inverse of the given (direct) function $y = f(x)$ as $f^{-1}(x)$. Also, D , R indicate the domain and the range, respectively.

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|---|---|
| (a) $f^{-1}(x) = \frac{2 \ln x + 5}{\ln x - 1}$ | $D_{f^{-1}} = (0, \infty) \setminus \{e\}, R_{f^{-1}} = \mathbb{R} \setminus \{2\}$ |
| (b) $f^{-1}(x) = \sqrt{x^2 - 1}$ | $D_{f^{-1}} = [1, \infty), R_{f^{-1}} = [0, \infty)$ |

$$(c) \quad f^{-1}(x) = \sqrt{e^x - 5}$$

$$D_{f^{-1}} = (\ln 5, \infty), \quad R_{f^{-1}} = (0, \infty)$$

$$(d) \quad f^{-1}(x) = \frac{1}{2}(1 - \sqrt{5 + 4x})$$

$$D_{f^{-1}} = \left[-\frac{5}{4}, \infty\right), \quad R_{f^{-1}} = \left(-\infty, \frac{1}{2}\right]$$